

Register Set Design

Abdullah All Mamun Siam
Lecturer, DIU

Register

Register is a very fast computer memory, used to store data/instruction in-Execution.

A Register is a group of flip-flops with each flip-flop capable of storing one bit of information. An n-bit register has a group of n flip-flops and is capable of storing binary information of n-bits.

There are several types of registers:

1. General Purpose Registers
2. Data Registers
3. Address Registers
4. Condition Codes/FLAG Registers

Register

There are several types of registers:

1. General Purpose Registers: General-purpose registers can be assigned to a variety of functions by the programmer.
2. Data Registers: Data registers can be used only to hold data and cannot be employed in the calculation of an operand address.
3. Address Registers: Address registers can be general purpose or can be devoted to a particular addressing mode.
4. Condition Codes/FLAG Registers: Condition codes/FLAG register which is at least partially visible to the user holds condition codes (also referred to as flags).

D Flip-flop

D	Q(t + 1)
0	0
1	1

Fig: D Flip-flop truth table

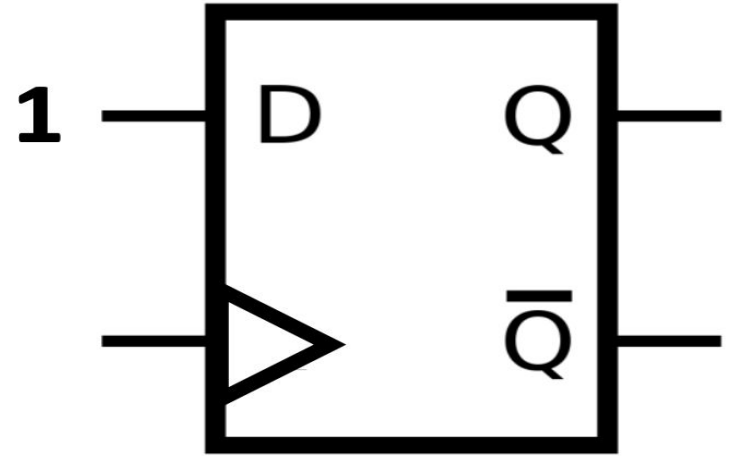
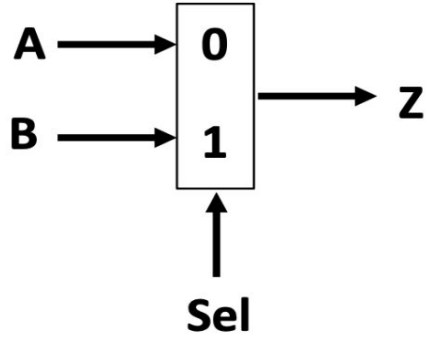


Fig: D Flip-Flop

1-bit Register Design



2 to 1 MUX

Sel	Z
0	A
1	B

$$Z = \overline{\text{Sel}} \cdot A + \text{Sel} \cdot B$$

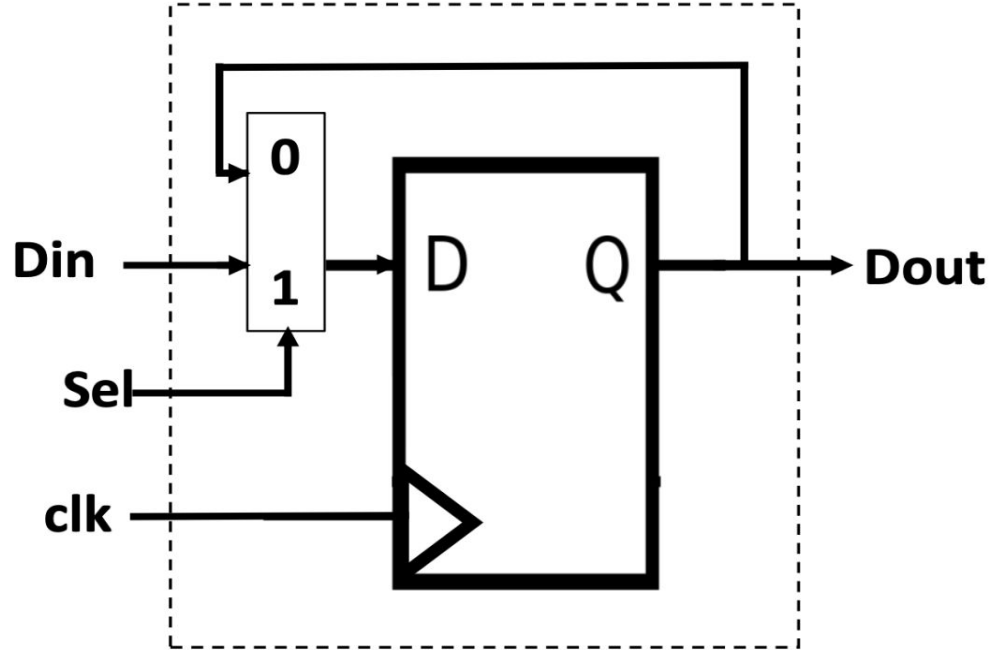


Fig: 1-bit Register

1-bit Register Design

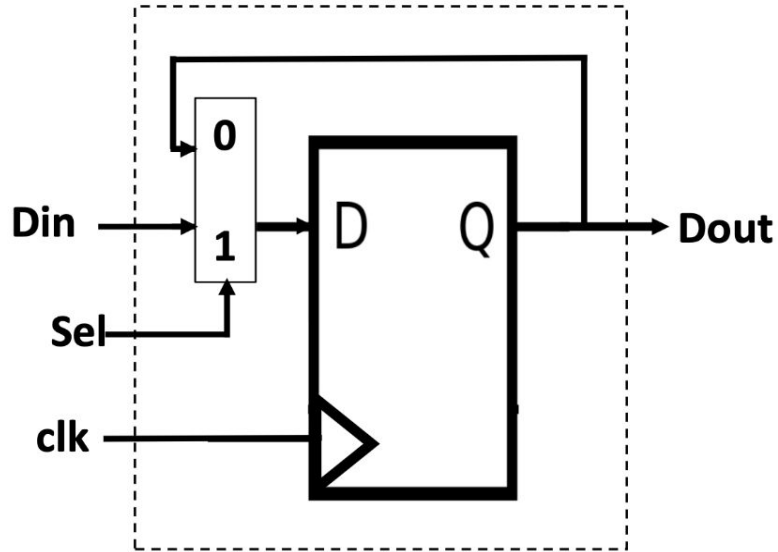


Fig: 1-bit Register

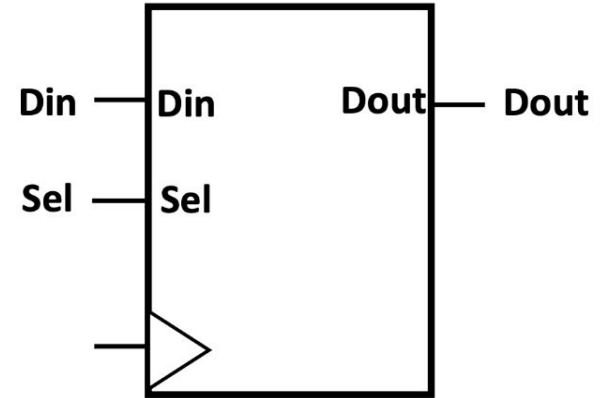
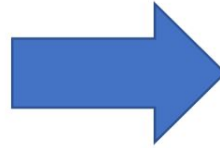


Figure: 1 bit register chip

2-bit Register Design

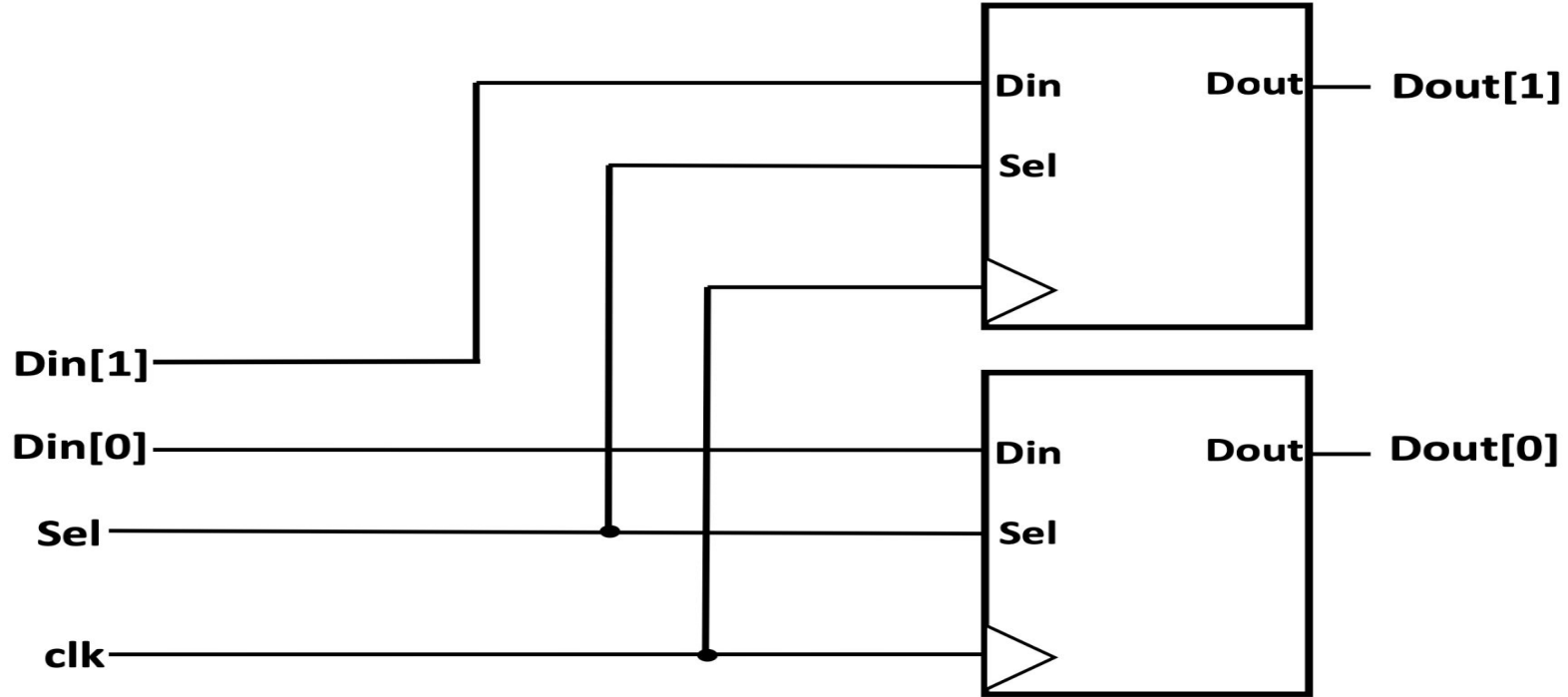


Figure: 2 bit Register

2-bit Register Design

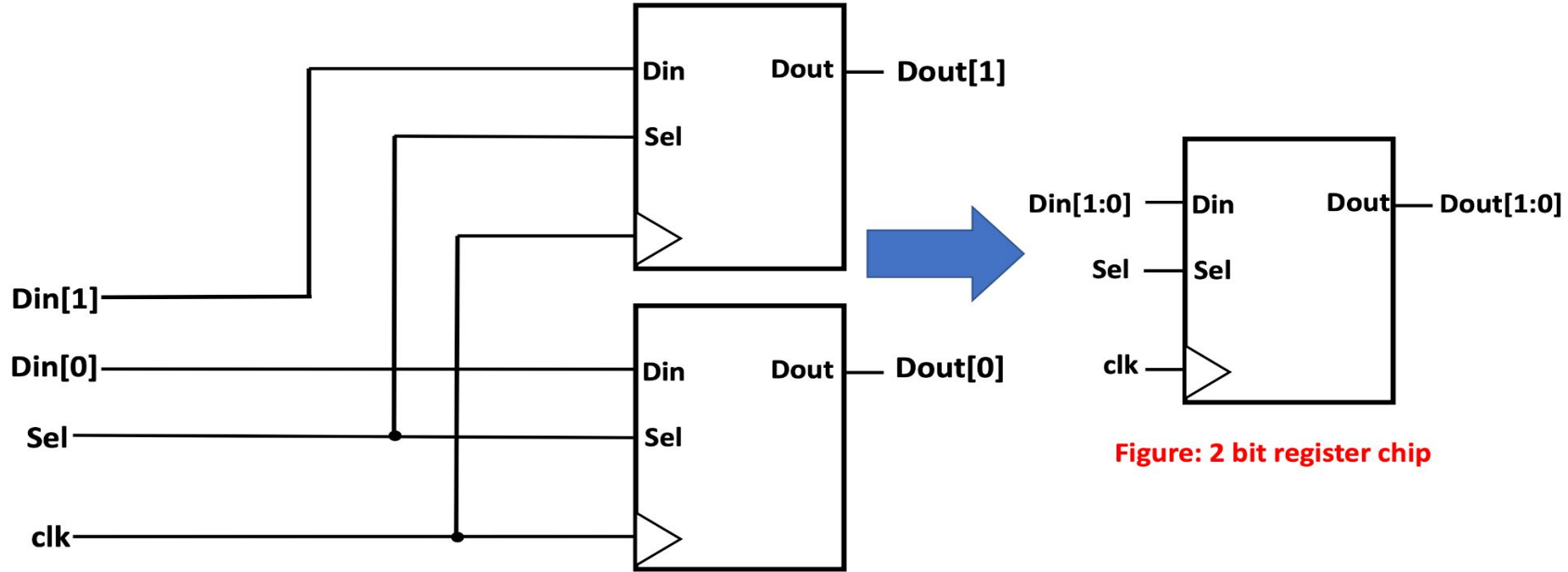


Figure: 2 bit register chip

Figure: 2 bit Register

4-bit Register Design

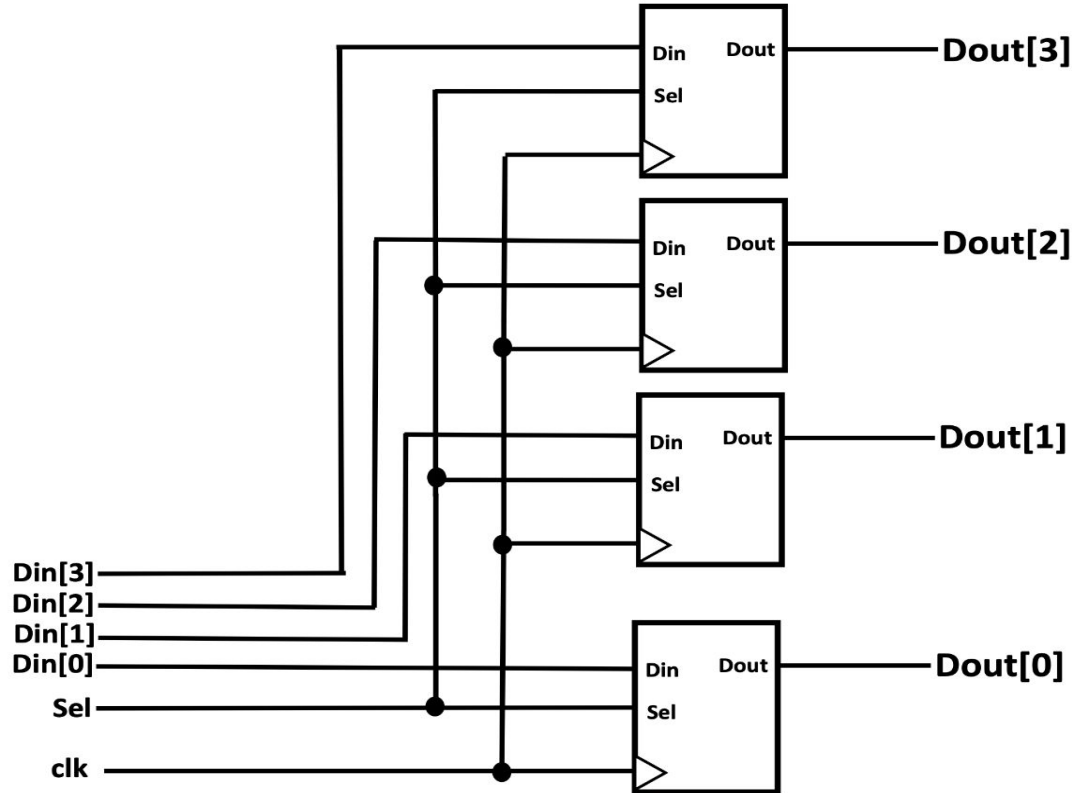


Figure: 4 bit Register

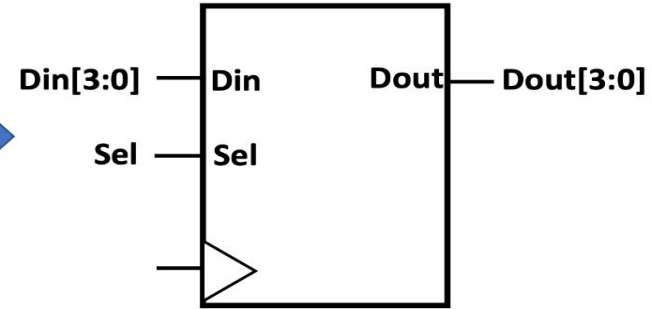
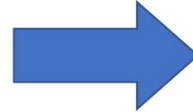


Figure: 4 bit register chip